

Social Media Privacy Guard: Context-Aware AI Anonymization

Project Guide

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Abstract

This project, Social Media Privacy Guard: Context-Aware AI Anonymization, aims to enhance digital privacy protection for social media users through intelligent and automated technology. The system detects sensitive information such as faces, license plates, identity documents, house numbers, and personal text that may be unintentionally exposed in images and videos. Unlike traditional tools that apply blanket blurring and reduce image quality, the proposed system analyses the context of the scene and selectively anonymizes only relevant sensitive elements. Advanced deep learning models and OCR techniques enable accurate detection, while user-controlled privacy rules allow customizable protection. The system also incorporates a Privacy Risk Score to assess exposure levels and operates with a privacy-first approach through local processing. By combining context awareness, intelligent decision-making, and quality preservation, the proposed model promotes secure, scalable, and user-friendly digital content sharing.

Key Features of the Proposed System

1. **Detection of Sensitive Objects:** Uses deep learning models (YOLO, CNN-based face detection) and OCR to identify faces, license plates, documents, and sensitive text.

2. **Context-Aware Selective Anonymization:** Instead of blindly blurring everything, the system analyses context and decides what should be anonymized.
3. **Main Subject vs Background Identification:** Preserves important content while protecting only sensitive background information.
4. **Privacy Risk Score (0–100):** Measures exposure level based on number, type, and visibility of sensitive objects.
5. **User-Controlled Privacy Rules:** Allows users to choose what to blur (faces, plates, documents, text, etc.).
6. **Explainable Highlight Mode:** Displays labelled bounding boxes before applying blur for transparency.
7. **Adjustable Blur Strength & Before–After Comparison View:** Maintains HD quality while enabling comparison of original and protected image.
8. **Privacy-First Local Processing:** Ensures no cloud storage and secure on-device processing.

SDG Goals

1. **SDG 9 – Industry, Innovation and Infrastructure:** Promotes innovative AI-based privacy solutions for digital platforms.
2. **SDG 4 – Quality Education:** Encourages awareness and responsible digital content sharing practices.

Advantages of Proposed System

- Context-aware selective anonymization
- Measurable privacy assessment through risk scoring
- Enhanced user customization and control
- Reduction in manual editing effort
- Preservation of image quality and aesthetics
- Real-time automated processing
- Secure local privacy-first architecture

Conclusion

The proposed system transforms traditional blanket blurring tools into a smart, context-aware, and privacy-first AI solution. By integrating deep learning detection, OCR, contextual reasoning, and adaptive anonymization techniques, it ensures secure, efficient, and user-friendly privacy protection. The model establishes a strong foundation for future AI-driven digital safety systems and responsible content sharing platforms.

POs Mapped: PO1, PO2, PO3

PSOs Mapped: PSO2, PSO3